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Research on Multi-unit Auction Theory and Application

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Abstract. Auctions usually involve the sale of many homogeneous or heterogeneous objects. Spectrum, treasury, and electricity auctions are examples. This kind of auctions is so-called multi-unit auctions. In order to know the research history and the newest research trends for the study on multi-unit auctions, this paper presents widespread theoretical and applications problems of multi-unit auctions. The multi-unit auction models are divided into two kinds (i.e. sequential auction and simultaneous auction). Assumptions, main ideas, and conclusions of literature related to these two kinds of models are reviewed and commented.

Introduction

In the practical auctions, such as spectrum auctions, treasury auctions and electricity auctions, which usually involves the sale of many homogeneous or heterogeneous objects. We called this kind of auctions as multi-unit auctions. Recently, there has been considerable attention paid to multi-unit auctions, and multi-unit auctions are widely used to sell identical, or quite similar goods. Sometimes sellers auction all goods together in a single round, but other times they auction them separately in a sequence of rounds. For example, cattle, fish, vegetables, timber, tobacco, and wine typically are sold in sequence, while government securities and mineral rights are sold in a single round. When using sequential auctions, the seller must decide what information to release after each round of bidding[1]. From this angle, multi-unit auctions can be divided into two kinds, i.e. sequential auctions and simultaneous auctions. Based on this standard, in the following discussion, the multi-unit auction models are divided into sequential auction models and simultaneous auction models.

In this paper, we reviewed and commented the assumptions, main ideas, and conclusions of literature related to the sequential auction models and simultaneous auction models. We will explore and clarify several important questions, namely, do sequential sales raise the seller's revenue, or is revenue maximized in a simultaneous auction? What is the newest research trends for the study on multi-unit auctions? What the challenging questions have given rise in the extension from traditional auctions to multi-attribute auctions?

Simultaneous auction

On the simultaneous auctions for homogeneous substitute goods, Maskin and Riley [2] analyzed the case of unit demands and continuously divisible goods, he extended the concepts and conclusions of single-item auctions to the multi-unit auctions, and proved the General Revenue Equivalence Theorem under the conditions of the homogeneous multi-unit goods and unit-demand. Wilson [3] was the first to consider uniform price auctions of divisible goods. He shows, in several settings, that there exists an equilibrium in which the asset is sold for half of its value. In many of these equilibria, bidders are able to maintain non-competitive quotes. This results in low revenues for the seller. This conclusion presents the problems or challenges. on the study of combinatorial auctions and homogeneous goods auctions.

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Kayhan et al. [4] studied a new ross-bidding strategy in online simultaneous auctions. They found that cross-bidding behavior has two important determinants: (i) market liquidity, defined as the number of simultaneous auctions that are available to bidders at any given instant, and (ii) number of experienced bidders. Apparently, as the number of simultaneous auctions increased, more bidders engaged in cross-bidding and this tendency increased with greater numbers of experienced bidders in the market. They also pointed out that the cross-bidding strategy may be limited by certain boundary conditions. For example, it may work if only a small portion of bidders, but not the entire population, cross-bid. If all bidders in the market monitor and bid on competing simultaneous auctions, the informational advantage held by cross-bidders is lost.

Wang and Zender [5] studied the divisible good simultaneous auctions, he derived equilibrium bidding strategies in for asymmetrically informed risk neutral and risk averse bidders when there is random noncompetitive demand, and pointed out that when the bidders' information is symmetric, the strategic aspects of bidding imply that there always exist equilibria of a uniform-price auction with lower expected revenue than provided by a discriminatory auction. Hoever, when bidders are risk averse, there may exist equilibria of the uniform-price auction that provide higher expected revenue than a discriminatory auction.

Damianov et al. [6-7] compared the uniform price and the discriminatory auction in a setting of supply uncertainty, where uncertainty is caused by the interplay of two factors: the seller's private information about marginal cost and the seller's incentive to sell the profit-maximizing quantity, given the received bids. They concluded that in every symmetric mixed strategy equilibrium, bidders submit higher bids in the uniform price auction than in the discriminatory auction. In the two-bidder case, this result extends to the set of rationalizable strategies. As a consequence, they found that the uniform price auction generates a higher expected revenue for the seller and a higher trade volume. Furthermore, they experimentally investigated whether a uniform or a discriminatory price auction is better for the seller in this setting.

Furthermore, Genc [8] compared results for discriminatory auctions to results for uniform-price auctions when suppliers have capacity constraints. He has a pretty good understanding of what equilibrium results look like for the uniformprice auctions. But an unresolved problem is what happens when a discriminative auction is run and suppliers have capacity constraints. They showed that payments made to the suppliers in the unique equilibrium of the discriminatory auction can be less than the payments in the uniform-price auction, depending on which uniform-price auction equilibrium is selected. For the high demand and/or low excess capacity cases, we characterize also mixed strategy supply function equilibrium under the discriminatory auction.

On the simultaneous auctions for heterogeneous imperfect substitute goods, Bapna et al. [9] studied settings where a number of sellers simultaneously offer vertically differentiated Vickrey auctions. They found that low risk type bidders tend to crowd on the highest auction and will pay a premium for the certainty it offers, whereas high risk type bidders fail to appropriately adjust for risk associated with the lowest auction, leading to overbidding. These lead to an interesting focal anomaly whereby bids are concentrated on the highest and lowest auctions, bypassing intermediate auctions. for imperfect substitute goods to unit-demand buyers. Hsieh [10] considered a winner determination problem for combinatorial reverse auction and study how to support the bidders' decisions without explicitly disclosing the bids of others. He proposed an information revelation scheme for a buyer to guide the sellers to generate potential winning bids to minimize the overall cost.

Andersson and Wilenius [11] proved that in many cases, a first-price sealed-bid combinatorial auction gives higher expected revenue than a sealed-bid simultaneous auction. This is the first theoretical evidence that combinatorial auctions indeed generate higher revenue, which has been a common belief for decades. They also provided an upper bound on the expected revenue for simultaneous auctions and a lower bound on combinatorial auctions, and derived an asymptotic result, proving that as the number of bidders approach infinity, expected revenue of the combinatorial auction will be higher than that of the simultaneous auction.

Chiou et al. [12] considered that although there are some studies on multi-unit auctions, most of them restrict attention to the case of one market so they studied an auction game in which simultaneous English auctions take place at several separate markets, and each market has multiple identical units of a good to sell. A bidder who knows her own private valuation of the good and the valuation probability distribution over other bidders has to select one market to participate and each bidder needs at most one unit of the good. They also analyzed the bidders mixed-strategy Bayesian Nash equilibrium and develop an algorithm to solve the equilibria of these games with finite types and finite markets, and examined the expected prices and the average surplus gained by the agents with a successful bid among markets.

Sequential auction

Sequential auction is another important part of the study on multi-unit auctions. There has been considerable attention paid to sequential auctions. Milgrom and Weber [13] compared the efficiency for a variety of sequential auction mechanisms under some general assumptions. Concretely, they developed a model of competitive bidding in which the winning bidder's payoff may depend upon his personal preferences, the preferences of others, and the intrinsic qualities of the object being sold. They concluded that the English auction generates higher average prices than does the second-price auction. Also, when bidders are risk-neutral, the second-price auction generates higher average prices than the Dutch and first-price auctions. They also showed that in all of these auctions, the seller can raise the expected price by adopting a policy of providing expert appraisals of the quality of the objects he sells. Kannan [14] showed that in a sequential-auction setting, the expected-winning prices are shown to decline across two stages when all bids are revealed between the stages. The prices decline because bidders desire to hide their private valuation information. The hiding also leads to inefficient allocations. Pitchik [15] studied a budget-constrained, private-valuation, sealed-bid sequential auction with two incompletely-informed, risk-neutral bidders in which the valuations and income may be non-monotonic functions of a bidder's type. They concluded that multiple equilibrium symmetric bidding functions may exist that differ in allocation, efficiency and revenue. The sequence of sale affects the competition for a good and therefore also affects revenue and the prices of each good in a systematic way that depends on the relationship among the valuations and incomes of bidders.

Furthermore, some scholars have compared and analyzed the advantages and disadvantages between the sequential auctions and simultaneous auctions. Said [16] analyzed a dynamic market in which buyers compete in a sequence of private-value auctions for differentiated goods. They considered the use of second-price auctions for selling these objects, and showed that there exists a unique symmetric Markov equilibrium in this market. In equilibrium, buyers shade their bids down by their continuation value, which is the (endogenous) option value of participating in future auctions. Mezzetti et al. [1] studied sequential and single-round uniform-price auctions with affiliated values, and derived symmetric equilibrium for the auction in which k_1 objects are sold in the first round and k_2 in the second round, with and without revelation of the first-round winning bids. Then concluded that when the first-round winning bids are announced, they also identify a positive informational effect on the second-round price. Total expected revenue in a sequential uniform auction with winning-bids announcement may be greater or smaller than in a single-round uniform auction, depending on the model's parameters. Feng and Chatterjee [17] studied a uniform-price auction, in which bidders demand no more than one item, and have independent, private values to find out two important problems, i.e., under what circumstances will a seller be better off selling the inventory sequentially, rather than selling it all in a single auction? Does it benefit the seller to reveal how many auctions there will be? They showed that even when all buyers are present at the beginning of the auction, and when both the seller and the buyers are impatient, a sequential sale can still benefit the seller as it stimulates competition among forward-looking bidders. However, this results depends on: (1) the bidder competition intensity, which is characterized by the difference between the number of

items available and the number of buyers; and (2) the impatience levels of the auctioneer and the bidders. They also showed that when there is uncertainty about the quantity supplied, a high-inventory auctioneer may benefit from using sequential auctions by pretending to be a low-inventory one initially.

These are all the important research results in multi-unit auctions theory in the past few years. However, most of these results are obtained based on some simple and special conditions, such as unitary demand for every bidder's valuation follows uniform distribution, the bid price of bidders are discrete, the bidders are symmetrical and so on [18,19]. Compared with the actual auction there are still some deficiencies, for example, (1) The fixed information structure and the determinate auction goods are the common assumptions in the most auction models. But in practice, many to many auctions, especially online auctions, since the auctioneers and the bidders join the auction randomly, the auction information is difficult to achieve ahead, and the asymmetric information structure will be changed as the disclosure of information gradually. (2) Information asymmetry between buyers and sellers is the core of the traditional auction theory, but there are a lot of information asymmetry between buyers and buyers or sellers and sellers in many to many auctions. (3) "Gives the dominant offer" may be the unique strategy for the bidder in the above study. In fact, because of complex structure for the information asymmetry, there exists strategies choice problems of information disclosure, and dominant offer strategy is the rational choice result for the bidders.

Prospects for Future Research

Up to now, there are abundant research results of theory and applications on the multi-unit auctions. It has developed a mature theoretical framework, and its applications have become increasingly widespread. However, there are still many important issues to be resolved.

Traditional theory research on multi-unit auctions is often aimed at some characteristics of specific forms of the auctions. So it is difficult to apply the auction models to the policy-making for government's large-scale resource allocation problem. How to really make the auction theory becomes the mainstream of economics, and how to use auction theory to guide the reality of commercial activities, and guide the formulation of government policy, it is still further worthy study problems for the researchers.

With the multi-unit auctions in their daily lives become increasingly commonplace, it is more and more difficult to simulate the psychological value by using the independent private valuation models. Facts have shown that correlation valuation models and common valuation models are two necessary models to simulate the psychological value of the bidders. However, there are very few studies on these two types of valuation model. This is also a deficiency in the practical application of auction theory. In addition, the existing literature on collusion of bidders are limited, especially, the reasonable and instructive results on there collusion is still less.

From the perspective of decision sciences, the players' game in the auction is a very complex decision-making process, the trading strategy should be further studied. Both the diversity of trading activity and market uncertainty are two difficulties. From the application point of view, most of the present studies done in a laboratory environment, but the theoretical models require a lot of of data support. This makes the most conclusions of auction theory is predictable, and have some differences to the actual situation.

The game theory as the main tool to develop auction theory has been shown to have important practical significance. Growing number of scholars have used static and dynamic game analysis methods to design the optimal auction mechanisms. However, game theory is aiming mainly at the symmetry equilibrium. However, most auctions in reality is clearly asymmetrical. Once back to full reality of asymmetric auction world, the game's solution will become extremely complex and diverse. Auction theory gives a lot of optimal mechanisms for the application, but these mechanisms should depend on the unknown distribution of bidder valuations. In addition, the auction model is still in the

existing partial equilibrium analysis phase, and the price system in coordinating the different aspects involved in the role of the main acts only in the general equilibrium system can be fully understood. Therefore, how to make the auction models into the framework of general equilibrium analysis, it is also a problem to be solved.

In e-commerce environment, multi-attribute online auctions did not receive due attention. The multi-attribute auctions are applied by many multinational companies in the online procurement. However, the basic theory study is lagging behind. It can not be said to be very puzzling. In this field, how to design a simple and effective optimal auction model is still the focus of future research for some time. Multi-attribute online auctions are bound to affect the restructuring of the entire enterprise supply chain, it will greatly shorten business procurement cycle and procurement cost savings.

As the continued innovation in practice, some new auction models are gradually developed. How to adapt to innovation in practice, and establish a new auction system, which are important research topics. In addition, it is not limited to the auction itself, while also taking into account the environment outside the auction is an important extension direction of auction theory.

Moreover, there are many challenging issues must be studied further for the auction theory. For example, how the auctions affect the performance of the enterprise and the way people live, how the resale market affect the auction income and auction efficiency; how the endogenous bidders affect the auction performance, and so on.

Conclusions

This paper reviewed and commented the literature related to the multi-unit auctions, and presents widespread theoretical and applications problems of multi-unit auctions. From these work, we can know the research history and the newest research trends for the study on multi-unit auctions clearly. In order to guide the readers study the multi-unit auctions deeply, we give some valuable prospects for the future research.

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